

PELVIS & LOWER LIMB FRACTURES

PELVIS

- Pelvic ring = 2 innominate bones + sacrum; anterior = **symphysis pubis**, posterior = **SI joints**. Stability from bone + ligaments (sacroiliac, sacrospinous, sacrotuberous, symphyseal).
- **Serious injuries**: soft-tissue, **blood loss (2–3 L)**, sepsis, ARDS; **mortality ~10%** if soft tissue involved.
- **Assessment**: ABCD first; severe pain, swelling/bruising (perineum/scrotum), urinary extravasation signs, perineal widening = disruption. **Blood at external meatus = urethral rupture → do NOT catheterize**. Rectal exam, sciatic nerve check.
- **Imaging**: AP pelvis (all polytrauma) + inlet/outlet/oblique views; CT for complex; urethrogram/cystogram.

Isolated fractures

- **Avulsion** (sportsmen): AIIS (rectus femoris), ASIS (sartorius), pubis (adductor longus), ischial tuberosity (hamstrings) → rest.
- **Direct** (ischium/iliac blade) → bed rest. **Stress** (osteoporotic, pubic rami, painless) → isotope scan.

Pelvic ring fractures (disrupted at 2 points)

- **AP compression (Open book)**: frontal collision; pubic rami fracture or symphysis disruption (**>2cm = unstable**), SI disruption.
- **Lateral compression (closed/buckle)**: side impact; pubic rami + SI/sacral fracture.
- **Vertical shear**: fall on one leg; innominate displaced vertically; **severe, unstable, organ injury, retroperitoneal haematoma**.
- **Management**: ABC, treat shock, **rapid pelvic binder/clamp**, laparotomy if needed; urologist for bladder/urethra. Stable open book (<2.5cm) → binder; **unstable open book (>2.5cm) → external fixation / symphyseal plate**; vertical shear → ext fix + SI screws.

Acetabular fractures

- Femoral head driven into pelvis (side blow / dashboard). Types: **wall** (with hip dislocation), **anterior column** (ilio-pectineal line, not weight-bearing), **posterior column** (ilio-ischial line, weight-bearing, post hip dislocation), transverse, complex.
- Exclude sciatic/femoral nerve injury. **Non-op** if <3mm / not weight-bearing / elderly → traction 6 wks. **ORIF** if displaced/unstable/retained fragment. Complications: AVN, sciatic nerve, heterotopic bone, OA.

HIP DISLOCATION

- **Posterior = 90% (commonest)**; dashboard injury (hip+knee flexed). **Leg: shortened, adducted, internally rotated, flexed**. Check **sciatic nerve**.
- **Thompson-Epstein**: I (none/small chip), II (large wall), III (comminuted wall), IV (column), V (femoral head).
- **Treatment: URGENT reduction under GA within 6 hours** (traction + flex hip/knee 90° + rotate → clunk); post-reduction CT. Type I → traction; II → ORIF wall; full weight-bearing after 12 wks.
- **Complications: sciatic nerve 10–20%**, vascular (superior gluteal a.), **AVN (10% if immediate reduction, 40% if delayed)**, myositis ossificans, OA. **Early reduction ↓ AVN**.

FEMORAL NECK — INTRACAPSULAR

- Females, 7th–8th decade; osteoporosis, DM, alcohol. Fall on trochanter or twisting. **Leg externally rotated + short.**
- Levels: subcapital, transcervical, basal.
- **Garden classification:** I (incomplete, impacted/valgus), II (complete, undisplaced), III (complete, partial displacement, varus), IV (complete, full displacement).
- **Poor union** (interrupted blood supply to proximal fragment, synovial fluid, no periosteum) → **no role for conservative treatment.**
- **Treatment:** <70 yrs → **closed reduction + internal fixation** (cannulated screws / DHS) — perfect reduction + rigid fixation. >70 **low-demand** → **hemiarthroplasty (bipolar)**. THR: >50 active/healthy, Garden III–IV, OA, non-union, metastatic.
- **Complications:** **mortality 20% at 4 months; AVN femoral head**, non-union, OA.

INTERTROCHANTERIC — EXTRACAPSULAR

- **Older patients, osteoporotic women (F:M 6:1); unite easily, rarely osteonecrosis** (extracapsular). Fall on trochanter. **Leg short + externally rotated.**
- **Kyle classification** I–IV (I undisplaced → IV severe comminution/reverse oblique/subtrochanteric).
- **Treatment: ALWAYS surgery** (early mobilization). **Dynamic hip screw (DHS) = stable; proximal femoral nail (PFN) = unstable/reverse oblique/subtrochanteric;** THR if severe comminution/OA/failed fixation.
- **Prognosis: 25–30% mortality in 1st year** (worse: male, intertrochanteric>intracapsular, op delay >2 days, >80 yrs, ≥2 comorbidities). Malunion common (varus).

FEMORAL SHAFT

- Strong muscles → severe displacement. **High energy (young, RTA — 1–2 L blood loss), low energy (elderly/pathological), child abuse.**
- Types: spiral (FFH), transverse/oblique (direct), comminuted, segmental.
- **Clinical: ATLS** (shock, other injuries); examine neurovascular, hip, pelvis. Imaging: 2 views, 2 joints; CT for concomitant (neck, acetabulum, plateau).
- **Emergency:** Thomas splint / skeletal traction.
- **Definitive:** children <5 → traction + spica; 5–10 (<50kg) → flexible IM nail; **adults** → **locked IM nailing (standard)**; plating for distal/intra-articular/ipsilateral neck/growing child; **ext fix for compound type III/multiple/vascular**; pathological → IM nail + cement.
- **Complications:** shock, **fat embolism + ARDS** (multiple fractures), thromboembolism, infection; malunion, delayed/non-union.

DISTAL FEMUR (supracondylar)

- **Bimodal:** young (high energy) / elderly osteoporotic (low energy). Supracondylar, intercondylar (T/Y), comminuted.
- **Displacement: distal fragment into extension (gastrocnemius) → endangers POPLITEAL ARTERY;** varus (adductor magnus); shortening (hamstring/quads).
- **Clinical: examine distal pulses.** CT for intra-articular (Hoffa fracture).
- **Treatment:** non-op rare (non-displaced/non-ambulatory, hinged brace). Operative: **retrograde IM nail** (no intra-articular, osteoporotic), **locking plate** (comminuted), **dynamic condylar screw** (intra-articular no comminution).
- **Complications: popliteal artery damage** (even minimal displacement → explore/repair), stiffness, non-union, malunion.

KNEE DISLOCATION (emergency)

- High violence → cruciate + collateral rupture. Gross deformity. **Exclude popliteal artery + common peroneal nerve.**
- **Urgent reduction under GA**, back slab 15° flexion, monitor circulation 1 week. Open reduction if irreducible/open/vascular/obese.
- **Complications: popliteal artery 40–50% (immediate repair + fasciotomy)**, common peroneal nerve 25% (neuropraxia), instability, stiffness.

PATELLA

- Largest sesamoid; part of extensor mechanism. **Direct** (fall/dashboard → undisplaced/comminuted) or **indirect** (passive flexion + quads contraction → transverse displaced).
- **Test active knee extension** (intact = extensor mechanism OK); aspirate → haemarthrosis + fat droplets.
- **Treatment:** undisplaced + intact extensor → POP cylinder in extension. **Displaced transverse / extensor failure → ORIF tension band (2 K-wires + figure-of-8)**. Comminuted non-salvageable → patellectomy.

TIBIAL PLATEAU

- Periarticular; **male 40s (high energy), female 70s (falls)**. Varus/valgus + axial load (**bumper fracture**). Associated meniscal/ACL injury.
- **Schatzker classification:** I (lateral cleavage, young), II (cleavage + depression, older), III (pure central depression, osteoporotic), IV (medial condyle — peroneal nerve risk), V (bicondylar), VI (plateau + metaphysis-diaphysis dissociation).
- **Clinical: doughy haemarthrosis; neurovascular exam (Type IV → common peroneal neuropraxia); exclude compartment syndrome.** CT for articular depression.
- **Treatment:** I non-displaced → aspirate + cast; displaced → 2 cancellous screws. II/III → if depression <5mm conservative; severe → elevate + bone graft + buttress plate. IV → medial buttress plate. V/VI ("**Span, Scan, Fix**") → buttress plates or circular ext fixator (Ilizarov).
- **Complications:** compartment syndrome, neurovascular, blisters; stiffness, OA.

TIBIA/FIBULA SHAFT

- **Most common long bone fracture; subcutaneous → often compound (open).**
- **Twisting** → spiral, low energy, minimal soft tissue, bone penetrates skin from within. **Direct/angulatory** → transverse, high energy, comminution, more compound.
- **Always check for compartment syndrome.**
- **Treatment:** conservative (low energy, closed, pediatric) → MUA + full POP (midhigh→metatarsal), functional brace at 4 wks. **Operative: closed locked IM nailing = standard;** percutaneous elastic nail (children); ext fix (compound/infected/bone gap); locking plate (metaphyseal).
- **Complications:** vascular (proximal), **compartment syndrome**, infection (open), malunion, delayed/non-union (common — poor soft tissue), stiffness.

ANKLE (Malleolar / Pott's)

- Twisting with foot anchored. Pushed malleolus → **oblique**; pulled → **transverse**.
- **Danis-Weber (by fibular fracture level):** **A** (below syndesmosis, transverse lateral malleolus — internal rotation/adduction), **B** (at syndesmosis, oblique — external rotation, ± deltoid rupture), **C** (above syndesmosis — abduction, syndesmotom disruption, ± posterior malleolus).
- **Imaging:** AP, lateral, **30° oblique (mortise)** view — check medial joint space, syndesmosis (diastasis).

- **Treatment:** conservative if isolated non-displaced medial/lateral malleolus or posterior malleolus <25% / <2mm step → below-knee slab then cast 6–8 wks. **Operative (displaced A/B, all C): ORIF lateral malleolus first (most important)**, then medial (malleolar screws/tension band), syndesmotomic screw if unstable, posterior malleolus if large.
- **Complications:** vascular (if subluxation), wound breakdown; malunion, **non-union (medial malleolus)**, stiffness, OA.

HIGH-YIELD ONE-LINERS

- **Pelvis:** open book >2.5cm = unstable → ext fix; blood at meatus = urethral injury, don't catheterize; pelvic binder for shock.
- **Posterior hip dislocation** = 90%, leg flexed/adducted/internally rotated; reduce <6h; AVN risk (early reduction ↓).
- **Femoral neck (intracapsular)** = Garden classification; <70 → fixation, >70 → hemiarthroplasty; AVN risk; no conservative.
- **Intertrochanteric (extracapsular)** = DHS (stable)/PFN (unstable); unites well, no AVN; 25–30% 1-yr mortality.
- **Femoral shaft** = locked IM nail; watch fat embolism + blood loss.
- **Supracondylar femur + knee dislocation** → **POPLITEAL ARTERY**. Knee dislocation = 40–50% popliteal injury.
- **Patella** = test active extension; transverse displaced → tension band.
- **Tibial plateau** = Schatzker; bumper fracture; exclude compartment syndrome.
- **Tibia shaft** = commonest long bone, often open; IM nail; compartment syndrome.
- **Ankle** = Danis-Weber A/B/C; fix lateral malleolus first; mortise view.

PELVIS & LOWER LIMB FRACTURES — Revision Table

Bold = high-yield. Classifications + key management (proximal → distal).

PELVIS & ACETABULUM			
Topic	Key points		
Pelvic ring	2 innominate + sacrum (symphysis ant, SI joints post). Blood loss 2–3L, mortality ~10%. Blood at meatus = urethral rupture → no catheter		
AP compression (open book)	Frontal collision; symphysis disruption >2cm unstable → ext fix / symphyseal plate		
Lateral compression (closed book)	Side impact; pubic rami + SI/sacral; ext fix ± plate		
Vertical shear	Fall on one leg; innominate displaced; severe/unstable; ext fix + SI screws		
Isolated avulsion	AIIS (rectus), ASIS (sartorius), pubis (adductor), ischial tub (hamstrings) → rest		
Acetabular	Anterior column (ilio-pectineal, non-WB) / posterior column (ilio-ischial, WB, post dislocation) ; ORIF if displaced; AVN/sciatic/OA		
HIP & PROXIMAL FEMUR			
Injury	Classification	Treatment	Key
Posterior hip dislocation (90%)	Thompson-Epstein I–V	Urgent reduction <6h under GA	Leg flexed/adducted/IR ; sciatic 10–20%; AVN 10% (early) → 40% (delayed)
Femoral neck (intracapsular)	Garden I–IV	<70 → CRIF (cannulated screws/DHS); >70 → hemiarthroplasty; THR if active/OA	Leg short + ER; AVN risk; NO conservative; mortality 20% at 4mo
Intertrochanteric (extracapsular)	Kyle I–IV	DHS (stable) / PFN (unstable, reverse oblique) ; always surgery	F:M 6:1; unites well, no AVN; 25–30% 1-yr mortality
Femoral shaft	Spiral/transverse/comminuted	Adults → locked IM nail ; children <5 traction+spica, 5–10 flexible nail; ext fix if compound	ATLS; fat embolism + ARDS ; 1–2L blood loss
KNEE REGION			
Injury	Key points	Treatment	
Supracondylar femur	Distal fragment extension → POPLITEAL ARTERY ; bimodal	Retrograde IM nail / locking plate / DCS; check distal pulses	
Knee dislocation	EMERGENCY; popliteal artery 40–50% , peroneal nerve 25%	Urgent reduction + monitor circulation; repair + fasciotomy	
Patella	Test active extension ; haemarthrosis + fat droplets	Undisplaced → cast; displaced transverse → tension band ; comminuted → patellectomy	
Tibial plateau	Schatzker I–VI (IV = peroneal nerve; bumper fracture); exclude compartment syndrome	I screws; II/III elevate+graft+buttress; V/VI = Span/Scan/Fix (ext fix)	
LEG & ANKLE			
Injury	Classification	Treatment	
Tibia/fibula shaft	Commonest long bone, often OPEN ; twisting (spiral) vs direct (transverse)	Closed locked IM nail (standard) ; ext fix if compound; watch compartment syndrome	
Ankle (malleolar/Pott's)	Danis-Weber A (below syndesmosis), B (at), C (above)**	Non-displaced → cast; displaced A/B + all C → ORIF lateral malleolus FIRST , syndesmotic screw if unstable. Mortise view ; non-union = medial malleolus	